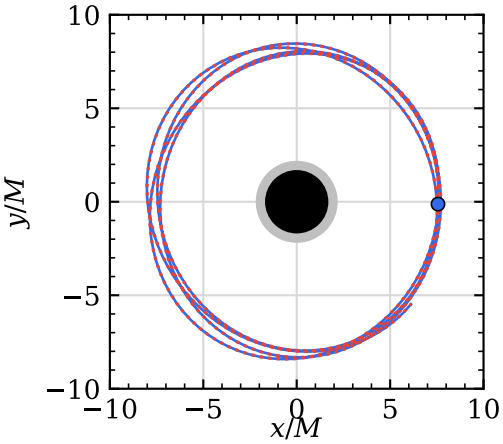
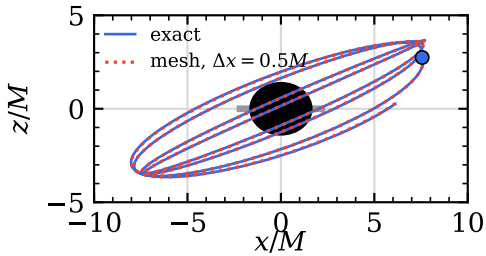
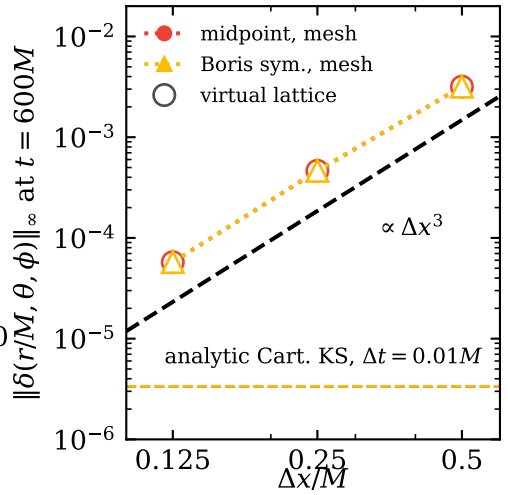
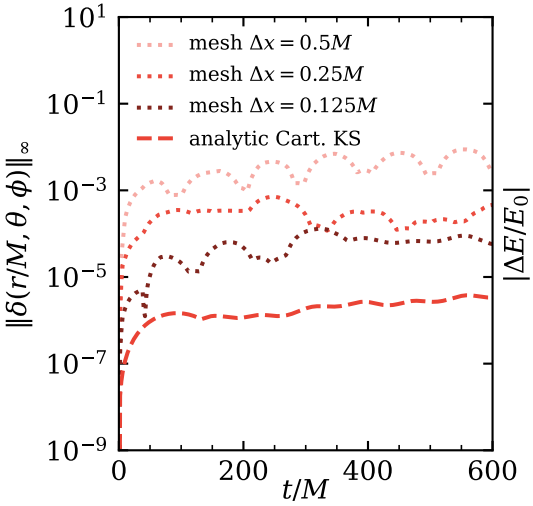
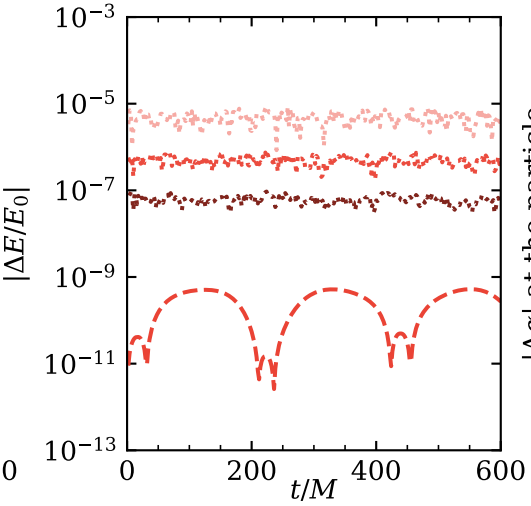
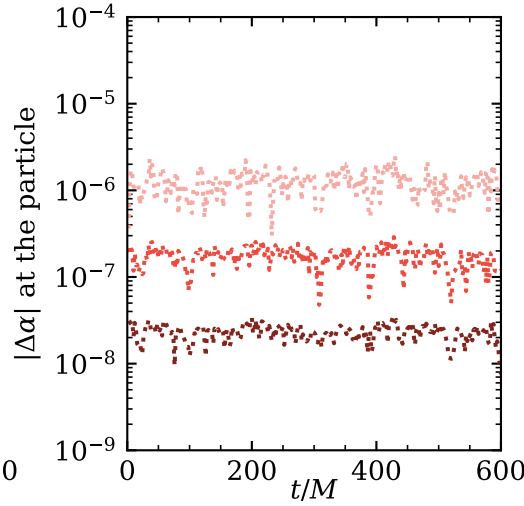

y/M
x/M


exact
mesh, Δx = 0.5M
z/M
x/M


midpoint, mesh
Boris sym., mesh
virtual lattice
∝ Δx³
analytic Cart. KS, Δt = 0.01M
‖δ(r/M, θ, φ)‖∞ at t = 600M
Δx/M


mesh Δx = 0.5M
mesh Δx = 0.25M
mesh Δx = 0.125M
analytic Cart. KS
‖δ(r/M, θ, φ)‖∞
t/M


|ΔE/E₀|
t/M


|Δα| at the particle
t/M